People of the Holocaust

Software Requirements Specification

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1 Revision History

Version	Date	Author	Description
1.0	October 4, 2026	Leyton Meadows	Initial software requirements specification.

2 Introduction

2.1 Purpose

This document defines the software requirements for the *People of the Holocaust* senior design project. It describes the required behavior of the public website, database, data processing pipeline, & testing processes.

2.2 Scope

People of the Holocaust is a web based historical research tool that provides person centered information from the *Encyclopedia of Camps and Ghettos*. The system will allow users to browse and search for people, places, and map locations. The system will eventually pull information from a database rather than CSV files - which is what it currently does.

The project includes a Python interpreter for processing source text, a database for storing structured records, and a React frontend for displaying said records to users.

2.3 Definitions and Acronyms

Term	Definition
ECG	Encyclopedia of Camps and Ghettos
Activity	A structured record showing an action or event associated with a person or place.
Interpreter	The Python data processing pipeline used to extract records from the ECG
Frontend	The React and Typescript web application used to show project data.
Database	Persistent data storage containing people, places, activities, & map records.

3 Overall Description

3.1 Product Perspective

The system has three main parts: a frontend website, a database, and a backend interpreter. The frontend gives the public a user interface. The database stores structured people, places, activities, and location records. The interpreter processes encyclopedia text and prepares records for storage in the database.

3.2 Product Functions

1. Display a public homepage explaining the purpose behind the project.
2. Allow users to search and browse people records.
3. Allow users to search and browse place records.
4. Display historical locations on an interactive map.
5. Store structured data in a database.
6. Process source material into structured records.

3.3 User Classes

1. Researchers and academics using the system for historical research.
2. Students using the system for learning and class projects.
3. General public users interested in the history behind the Holocaust.
4. Project maintainers responsible for the frontend, database, and interpreter.

4 Functional Requirements

ID	**Requirement**	**Priority**
FR-1	The system shall provide a public website for viewing *People of the Holocaust* data.	High
FR-2	The system shall not require users to create an account or log in to view public records.	High
FR-3	The website shall provide a homepage that explains the background and purpose of the project.	High
FR-4	The website shall provide a People page that pull person records from the database.	High
FR-5	The website shall allow users to search people by name.	High
FR-6	The website shall always allow users to select a person and view related details.	High
FR-7	The website shall provide a Places page that pulls place records from the database.	High
FR-8	The website shall allow users to search places by name.	High

FR-9	The website shall allow users to select a place and view related details.	High
FR-10	The website shall provide an interactive map of historical locations.	High
FR-11	The map shall display markers for places with valid latitude and longitude values.	High
FR-12	The map shall display place information when a marker is selected.	High
FR-13	The interpreter shall clean raw person names into structured person records.	High
FR-14	The interpreter shall retrieve and clean article text from the *ECG*.	High
FR-15	The interpreter shall identify sentences that contain known person names.	High
FR-16	The interpreter shall pull activity records from identified sentences when possible.	High
FR-17	The interpreter shall associate activity records with people and places.	High
FR-18	The interpreter shall write processed records to project database.	High
FR-19	The interpreter shall log article processing logs for later review.	High

5 Non-Functional Requirements

ID	**Requirement**	**Priority**
NFR-1	The website shall be usable in modern browsers.	High
NFR-2	The website shall pull public data from the database through API requests or database client.	High
NFR-3	The frontend shall be buildable using the project build command.	High
NFR-4	The interpreter shall run Python 3.11 or newer.	High
NFR-5	The system shall preserve historical meaning of extracted material.	High
NFR-6	The website shall remain usable when optional data fields are missing.	High
NFR-7	Database credentials and service keys shall not be exposed in frontend code or committed to version control.	High
NFR-8	The system shall respect source permissions.	High
NFR-9	The system shall be maintainable by future developers.	High

6 Data Requirements

Data Item	Description
People Records	records containing person identifiers, first names, last names, alternative names, titles, places, organizations, and designations.
Place Records	Records containing place identifiers, names, types, subtypes, countries, and location identifiers.
Activity Records	Records connecting people, places, actions, and supporting sentence/paragraph details.
Map Records	Records containing place names, latitude, longitude, type, subtype, and current country for map display.
Article Source Records	Records identifying encyclopedia articles processed by the interpreter.
Error Logs	Records of source articles or processing steps that failed during interpreter execution.

The database will be the primary source for website data. The CSV and JSON files will be used during development, testing, or migration, but they are not the intended long-term data source.

7 User Requirements

ID	User Requirement
USR-1	As a researcher, I want to search for a person by name so that I can find relevant historical records quickly.
USR-2	As a researcher, I want to search for a place so that I can find locations of interest.
USR-3	As a public user, I want to view an interactive map so that I can explore historical places geographically.
USR-4	As a public user, I want to access the website without an account.
USR-5	As a project maintainer, I want the interpreter to identify key sentences automatically so that data creation is more efficient.
USR-6	As a project maintainer, I want failed article processing attempts to be logged so that errors can be reviewed.

8 Constraints and Assumptions

8.1 Constraints

1. The frontend is implemented using React, TypeScript, and Vite.

2. The frontend is expected to retrieve structured records from the project database.
3. The interactive map depends on LeafLet and OpenStreetMap tiles.
4. The interpreter is implemented in Python and requires Python 3.11 or later.
5. Database access requires credentials that must not be committed to version control.
6. The system must respect source permissions for encyclopedia content.

8.2 Assumptions

1. The project has permission to use the required source material.
2. Database records are populated before frontend deployment.
3. records with missing optional fields should not prevent the website from loading.
4. A sentence may refer to more that one person.
5. Some extracted activity records may be incomplete when parsing cannot confidently identify an action.

9 Acceptance Criteria

ID	Acceptance Criterion
AC-1	The homepage loads and shows the project purpose and description.
AC-2	The People page loads person records from the database.
AC-3	Searching on the People page filters records based on entered text.
AC-4	Selecting a person displays that person's details and relationships.
AC-5	The Places page loads place records from the database.
AC-6	Searching on the Places page filters records based on entered text.
AC-7	Selecting a place shows that place's details and related activities.
AC-8	The map loads location records and shows markers for records with valid coordinates.
AC-9	Marker popups accurately show place information.
AC-10	The frontend build command completes successfully.
AC-11	The interpreter tests complete successfully.
AC-12	The interpreter logs failed processing attempts without stopping the full processing run.

10 Verification Plan

The requirements will be verified using build checks, manual website testing, and Python unit tests for the interpreter.

1. Run the frontend build process.
2. Manually verify homepage, navigation, People page, Places page, and map behavior.

3. Verify that people, place, activity, and map records correctly load from the database.
4. Verify that search inputs filter people and places correctly.
5. Verify that selected records display associated activities.
6. Run interpreter tests using `pytest`.
7. Review the interpreter error logs after processing articles.